

More Modules!

Ignition**Exercise 1.**

- (a) Suppose M is a free module. Show that M is projective.
- (b) Prove that if P is projective, then any exact sequence $0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$ splits on the right.
- (c) Any module is a quotient of a free module. A universal, but very wasteful, way to accomplish this would be to make F have one basis element e_m for every $m \in M$, and take the homomorphism $F \rightarrow M$ that sends e_m to m . Prove that if P is a module with the property that any exact sequence $0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$ splits, then P is a direct summand of a free module. That is, there is some module R so that $P \oplus R$ is free.
- (d) Show that if P is projective and $P = S \oplus T$, then S is also projective.
- (e) Combine parts (a) and (d) to conclude that if P is a direct summand of a free module, then P is projective.

Exercise 2.

- (a) Let $R = \mathbb{Z}/6\mathbb{Z}$, and let $M = \mathbb{Z}/3\mathbb{Z}$. Then M is an R -module under the action $r \cdot m = rm$ for $r \in R, m \in M$. Prove that M is projective, but not free.
- (b) Let $R = M_2(\mathbb{R})$, and let $M = \mathbb{R}^2$. Then M is a left R -module under the action $A \cdot v = Av$ for $A \in R$ and $v \in M$. Prove that M is a projective left R -module, but not a free left R -module.
- (c) Let R be a PID. What are all the finitely-generated projective R -modules?
- (d) Let M be a projective module over a commutative ring R , and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$ -module.

Exercise 3.

- (a) Prove that a free module is flat. (With a little more work, one can use this to show that a projective module is flat.)
- (b) Let R be an integral domain, and let M be a flat R -module. Prove that M has no torsion, i.e. there are no $m \in M \setminus \{0\}$ and $r \in R \setminus \{0\}$ so that $rm = 0$. (Hint: there is an exact sequence $0 \rightarrow R \xrightarrow{\times r} R \rightarrow R/(r) \rightarrow 0$.)
- (c) Let R be a PID. What are all the finitely-generated flat R -modules.
- (d) Let $S \subset R$ be a multiplicative set. Show that $S^{-1}R$ is a flat R -module. (Hint: $S^{-1}R \otimes_R M = S^{-1}M$.)

Exercise 4. Part (d) of the previous question implies that $\mathbb{Q}$ is a flat $\mathbb{Z}$ -module. Let $A = \mathbb{Z}^{\oplus \mathbb{N}}$ be a countably-generated free $\mathbb{Z}$ -module with basis $\{e_1, e_2, \dots\}$, and consider the homomorphism $f: A \rightarrow \mathbb{Q}$ which sends $e_i \mapsto \frac{1}{i}$. Prove that f is surjective, but there is no section $s: \mathbb{Q} \rightarrow A$. Hence, $\mathbb{Q}$ is not projective (nor indeed free).

Exercise 5. A (not necessarily commutative) ring R is **simple** if the only two-sided ideals of R are (0) and all of R .

A (not necessarily commutative) ring R is a **division ring** if for every nonzero $r \in R$ has a multiplicative inverse r^{-1} .

- (a) Prove that a division ring is simple.
- (b) Prove that a nonzero commutative simple ring is a field.
- (c) Let $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ be Hamilton's quaternions, which has multiplication defined by the relationships $i^2 = j^2 = k^2 = ijk = -1$ and $ij = -ji$. Show that $\mathbb{H}$ is a division ring.
- (d) Show that $M_2(\mathbb{R})$ is a simple ring but not a division ring.

Exercise 6. Similar to the above, a left (right) module M over a ring R is **simple** if the only left (right) submodules of M are 0 and all of M . (A non-simple ring can have simple modules.)

A module M is **cyclic** if there is some $m \in M$ so that the module generated by m is all of M .

- (a) Prove that a simple module is cyclic. (Hint: choose *any* nonzero $m \in M$.)
- (b) Let R be a ring, and let $\mathfrak{m}$ be a maximal left ideal of R . Show that $R/\mathfrak{m}$ is a simple left R -module.
- (c) Now let M be any simple left R -module. By part (a), $M = Rm$ for some $m \in M$. Show that $\text{Ann}(m)$ is a maximal left ideal of R . Conclude that every simple left module over R is of the form $R/\mathfrak{m}$. (Hints: (i) Suppose $I \supsetneq \text{Ann}(m)$, and take $i \in I \setminus \text{Ann}(m)$. What does part (a) say about im ? (ii) If $rm = m$, then $r - 1 \in \text{Ann}(m)$.)

Take Off

Exercise 7. (August 2020 Problem 1) On this problem, no justification is required; only the answer will be graded.

- (a) Give an example of a non-diagonalizable matrix in $\text{GL}_3(\mathbb{F}_p)$ (for some p).
- (b) Give an example of a nontrivial proper normal subgroup of $\text{GL}_3(\mathbb{F}_p)$ (for some p).
- (c) Give an example of a one-sided ideal (either left or right) in $M_2(\mathbb{C})$ which is not a two-sided ideal.
- (d) Give an example of a $\mathbb{Z}$ -module which is flat but not free.
- (e) Give an example of an exact sequence of abelian groups which is not split.

Exercise 8. (January 2024 Problem 4, notation altered) Let R be a commutative ring, and let A and B be commutative R -algebras.

- (a) Prove that if B is flat over R , and f is a non zero divisor in A , then $f \otimes 1$ is a non zero divisor in $A \otimes_R B$.
- (b) Prove that the flatness condition above is necessary by giving an example where part (a) fails if B is not flat over R .

Exercise 9. (January 2021 Problem 3) (This was on WS 3.)

- (a) Show that a projective module over an integral domain is torsion free.
- (b) Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:
 - (i) If M_1 and M_2 are projective, then so is M_3 .
 - (ii) If M_2 and M_3 are projective, then so is M_1 .

Exercise 10. (January 2021 Problem 2) Let k be a field, and let $R = M_2(k)$, the non-commutative ring of $n \times n$ matrices over k .

- (a) Give examples of a simple left R -module M and a simple right R -module N .
- (b) Can you find another simple left R -module M' that is not isomorphic to M ? (Either find one or explain why there is none such.)
- (c) Compute $\dim_k(N \otimes_R M)$.